

Ashish Gupta

Senior Software + AI/ML Engineer with 5+ years of experience building production-grade systems across Machine Learning, LLM/SLM applications, backend engineering, and cloud-native deployment. Hands-on experience in building RAG pipelines, integrating vector search, deploying AI services on AWS/GCP, and working with Llama-family models for domain-specific inference workflows.

Strong at model + system optimization, low-latency API design, scalable microservices, and mentoring engineers through debugging, architecture, and best practices.

PROFESSIONAL EXPERIENCE

Snapwork, Mumbai : Senior Full Stack Engineer - Platform Systems

SEPT 2024 - PRESENT

Built scalable APIs and background jobs in golang and C++, improved frontend performance by 45% using **React**, **TypeScript**, and **Webpack**, optimized deployments with custom shell scripts, and mentored junior developers in debugging, testing, and modular development. Utilize **Copilot** for **Flutter** app development.

Projects at Snapwork:

- Julius Baer** - Portfolio Management System (As Senior fullstack developer)
 - Migrated and optimized large Vue.js components to **React.js** + **TypeScript** following performance & memory best practices.
 - Built high-tracking backend services using **Golang** and **C++** for faster computation and stable execution.
 - Tech Stack:** React.js, TypeScript, golang, C++, Webpack.
- Medicard** - Flutter Modules (As Flutter dev and AI tools researcher)
 - Developed new Flutter-based modules for the MediGo App using Copilot-assisted workflows.
 - Researched and implemented Spec-Driven Development using **GitHub SpecKit**.
 - Tech Stack:** Flutter, Dart, Copilot, SpecKit.

CODE-B, Mumbai : Full Stack & Embedded Engineer → Senior Engineer

MAR 2021 - SEPT 2024

Designed ESP32 firmware in C++ and Rust, optimized sensor polling, and built Golang microservices with sub-100ms latency, along with Raspberry Pi I2C/GPIO drivers and lightweight edge inference tools integrating Llama-2 and other ML models. Developed a full-stack EMS (Energy/Inventory Management System) using AWS IoT Core (MQTT) and the MERN stack, enabling real-time power monitoring and remote device control via mobile and web applications.

Projects at Code-B:

- Selec IoT** - IoT Device Inventory & Energy Monitoring System (As Senior Fullstack and IOT dev)
 - Delivered full **IoT** ecosystem with **React.js** Dashboard, **React Native** App, and **Node.js** backend for real-time monitoring, alerts, and monthly billing.
 - Developed device firmware using **Arduino C++** / **ESP32**, and built **C++** & **Rust** microservices for high-speed parsing, alerting, and PDF generation.
 - Implemented real-time data streaming using sockets and **AWS IoT Core**, ensuring stable device-to-cloud communication using **Golang**.
 - Tech Stack:** AWS IoT, React.js (TS), React Native, Node.js, Golang, C++, Arduino C++, ESP32, Bootstrap, Docker, MongoDB.
- Penrose Systems** - MegaMatch Multiplayer Match Game (As Senior Fullstack dev managed team)
 - Built real-time multiplayer game using **Node.js**/**Nest.js**, **Socket.IO**, **Redis**, and **MongoDB**, with **Vue.js** + **Pixi.js** for gameplay and animations, deployed on **GCP**.
 - Designed scalable event-driven architecture enabling fast matchmaking, synchronized player movement, and smooth gameplay.
 - Tech Stack:** Node.js, Nest.js, Socket.IO, Redis, Pixi.js, Vue.js, MongoDB, GCP.
- Tech Agriculture** - Plant Care & Diagnostic System (As Senior fullstack and ML engineer)
 - Developed AI-based plant disease detection using **TensorFlow (ResNet50)** with a **React.js** portal and **Python** backend ML workflows.
 - Integrated **Llama-2** to generate remedy suggestions and care plans, deployed end-to-end on **GCP**.
 - Tech Stack:** TensorFlow, NLP, LLM, ResNet50, Llama-2, Python, FastApi, React.js, GCP.

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SOCIAL LINKS

- [Linked In](#)
- [Github](#)
- [HackerRank \(5 Stars\)](#)
- [X \(Twitter\)](#)

SKILLS

- Languages
 - ◆ C/C++
 - ◆ Golang
 - ◆ Rust
 - ◆ Python/FastApi
 - ◆ JavaScript/TypeScript
- Embedded & Edge
 - ◆ ESP32 (Arduino)
 - ◆ IOT
 - ◆ Raspberry Pi
 - ◆ Shell scripting
- AI/ML
 - ◆ TensorFlow
 - ◆ LLAMA (LLM)
 - ◆ Gemma-1
 - ◆ Transfer Learning
 - ◆ Pytorch Deep Learning, NLP
 - ◆ Langchain
 - ◆ Fine-tuning workflows (LoRA/PEFT exposure)
 - ◆ Vector DB integration (FAISS / Pinecone-like concepts)
- Cloud & DevOps
 - ◆ AWS
 - ◆ GCP
 - ◆ Docker
 - ◆ Cloud Run
 - ◆ GitHub/Gitlab
 - ◆ CI/CD: GitHub Actions/GitlabGi
 - ◆ Linux
 - ◆ Shell Scripting

ACHIEVEMENTS

- **Fast-Track Promotion:** Promoted from Junior to Senior Software Engineer at Code-B within 2 years based on performance and mentoring contributions.
- **5-Star HackerRank Badge:** Earned

Freelancing , Mumbai : *Software Engineer – Full Stack & Embedded Integration*

MAR 2020 - FEB 2021

- Built dashboards and backend tools using MERN stack and deployed them on GCP Cloud Run.
- Delivered clean UI/UX and mobile-responsive applications.
- Integrated microcontroller-based solutions with web platforms using MQTT and REST APIs.

EDUCATION

Gurunanak College, University of Mumbai, Mumbai – *Bachelor of Science in Information Technology*

JUNE 2019 - JULY 2022

Completed a three-year undergraduate program combining core subjects in mathematics, computer science, and emerging technologies. The curriculum emphasized software development, Embedded Systems, IOT, data structures, networking, databases, applied mathematics, and practical lab-based problem solving.

ADDITIONAL PROJECTS

- **Pet Identity System (Pawfriend)** - Built ESP32 QR sync system for street animals using Python+ React + GCP.
- **Raspberry Pi Live Stream with Object Detection** - Implemented real-time video streaming with object detection using OpenCV and Python on IOT Raspberry Pi. deployed lightweight models to run at the edge with minimal latency.

5-star rating in Problem Solving, showcasing strong algorithmic and analytical skills.

- **Media Recognition:** Pet Identity System project featured in media for its innovative approach to animal identification using ESP32 and cloud integration. [Media Link](#)

LANGUAGES

English – Full Professional Proficiency

Hindi – Native